KAIZEN MATHS

Lesson starter

Integration by Substitution and Parts

Mixed Method Selection

Name _____

S1

Choose an appropriate substitution for each integral.

$$\int 6(3x-7)^4\,dx$$

S2

Use integration by parts for each integral.

$$\int xe^{4x}\,dx$$

S3

Choose between substitution and integration by parts for each integral.

$$\int x\sin(2x)\,dx$$

S4

Choose an appropriate substitution for each integral.

$$\int 10e^{2x-4}\,dx$$

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

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Name _____

EA1

Use integration by parts for each integral.

$$\int \ln x \, dx$$

EA2

Choose between substitution and integration by parts for each integral.

$$\int \frac{2(6x+5)}{3x^2+5x+7} \, dx$$

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Mixed Method Selection

Name _____

EB1

Choose an appropriate substitution for each integral.

$$\int \frac{3(4x+1)}{2x^2+x+3} \, dx$$

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Mixed Method Selection

Name _____

EA1

Use integration by parts for each integral.

$$\int \ln x \, dx$$

EA2

Choose between substitution and integration by parts for each integral.

$$\int \frac{2(6x+5)}{3x^2+5x+7} \, dx$$

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Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Mixed Method Selection

Name _____

EB1

Choose an appropriate substitution for each integral.

$$\int \frac{3(4x+1)}{2x^2+x+3} \, dx$$

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Show your working

Teacher answers & working

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S1

Choose an appropriate substitution for each integral.

$$\int 6(3x-7)^4\,dx$$

Answer:

$$\frac{2}{5}(3x-7)^5+C$$

Working:

Let $u = 3x - 7$.

Differentiate: $\frac{du}{dx} = 3$, so $du = 3\,dx$ and $dx = \frac{1}{3}\,du$.

Substitute into the integral: $\int 6(3x-7)^4\,dx = \int 6u^4 \cdot \frac{1}{3}\,du$.

Simplify the constant: $\int 2u^4\,du$.

New power: $4 + 1 = 5$.

Integrate: $2 \cdot \frac{u^5}{5} + C$.

Simplify the coefficient: $2/5 = \frac{2}{5}$.

$\frac{2}{5}u^5 + C$.

Replace u with $3x - 7$: $\frac{2}{5}(3x-7)^5 + C$.

Teacher answers & working

Integration by Substitution and Parts · Mixed Method Selection

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S2

Use integration by parts for each integral.

$\int xe^{4x}\,dx$

Answer:

$\frac{xe^{4x}}{4} - \frac{e^{4x}}{16} + C$

Working:

Use integration by parts: $\int u\,dv = uv - \int v\,du$.

Choose $u = x$ and $dv = e^{4x}\,dx$.

Then $du = dx$ and $v = \int e^{4x}\,dx = \frac{e^{4x}}{4}$.

Substitute into the formula: $\int xe^{4x}\,dx = x \cdot \frac{e^{4x}}{4} - \int \frac{e^{4x}}{4}\,dx$.

Take out the constant: $\int \frac{e^{4x}}{4}\,dx = \frac{1}{4}\int e^{4x}\,dx$.

Integrate: $\frac{1}{4} \cdot \frac{e^{4x}}{4}$.

Simplify: $\int \frac{e^{4x}}{4}\,dx = \frac{e^{4x}}{16}$.

Therefore $\frac{xe^{4x}}{4} - \frac{e^{4x}}{16} + C$.

Teacher answers & working

Integration by Substitution and Parts · Mixed Method Selection

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S3

Choose between substitution and integration by parts for each integral.

$$\int x\sin(2x)\,dx$$

Answer:

$$-\frac{x\cos(2x)}{2}+\frac{\sin(2x)}{4}+C$$

Working:

Use $\int u\,dv = uv - \int v\,du$.

Choose $u = x$ and $dv = \sin(2x)\,dx$.

Then $du = dx$ and $v = -\dfrac{\cos(2x)}{2}$.

Substitute: $\int x\sin(2x)\,dx = -\dfrac{x\cos(2x)}{2} - \int\left(-\dfrac{\cos(2x)}{2}\right)dx$.

The two minus signs give a plus.

$\dfrac{1}{2}\int\cos(2x)\,dx = \dfrac{1}{2}\cdot\dfrac{\sin(2x)}{2}$.

Simplify: $\dfrac{1}{2}\int\cos(2x)\,dx = \dfrac{\sin(2x)}{4}$.

Therefore $-\dfrac{x\cos(2x)}{2}+\dfrac{\sin(2x)}{4}+C$.

S4

Choose an appropriate substitution for each integral.

$$\int 10e^{2x-4}\,dx$$

Answer:

$5e^{2x-4}+C$

Working:

Let $u = 2x - 4$.

Then $\dfrac{du}{dx} = 2$, so $dx = \dfrac{1}{2}\,du$.

Substitute: $\int 10e^{2x-4}\,dx = \int 10e^{u}\cdot\dfrac{1}{2}\,du$.

Simplify: $\int 5e^{u}\,du$.

Integrate: $5e^{u}+C$.

Replace u with $2x - 4$: $5e^{2x-4}+C$.

Teacher answers & working

Integration by Substitution and Parts · Mixed Method Selection

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EA1

Use integration by parts for each integral.

$$\int \ln x \, dx$$

Answer:

$x \ln x - x + C$

Working:

Use integration by parts because the integral contains a logarithm.

Choose $u = \ln x$ and $dv = dx$.

$\frac{du}{dx} = \frac{1}{x} \cdot 1 = \frac{1}{x}$.

$du = \frac{1}{x} \, dx$.

$v = \int 1 \, dx = x$.

Substitute into $\int u \, dv = uv - \int v \, du$.

$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx$.

Simplify the remaining integral: $\int 1 \, dx = x$.

Therefore $x \ln x - x + C$.

EA2

Choose between substitution and integration by parts for each integral.

$$\int \frac{2(6x+5)}{3x^2+5x+7} \, dx$$

Answer:

$2 \ln |3x^2 + 5x + 7| + C$

Working:

Let $u = 3x^2 + 5x + 7$.

Then $\frac{du}{dx} = 6x + 5$, so $du = (6x + 5) \, dx$.

The numerator contains $2(6x + 5)$, so the integral becomes $\int \frac{2}{u} \, du$.

Use $\int \frac{1}{u} \, du = \ln |u| + C$.

Therefore the integral is $2 \ln |u| + C$.

Replace u with $3x^2 + 5x + 7$: $2 \ln |3x^2 + 5x + 7| + C$.

Teacher answers & working

Integration by Substitution and Parts · Mixed Method Selection

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EB1

Choose an appropriate substitution for each integral.

$$\int \frac{3(4x+1)}{2x^2+x+3}\,dx$$

Answer:

$3\ln|2x^2+x+3|+C$

Working:

Let $u = 2x^2+x+3$.

Then $\frac{du}{dx} = 4x+1$, so $du = (4x+1)\,dx$.

The numerator contains $3(4x+1)$, so the integral becomes $\int \frac{3}{u}\,du$.

Use $\int \frac{1}{u}\,du = \ln|u|+C$.

Therefore the integral is $3\ln|u|+C$.

Replace u with $2x^2+x+3$: $3\ln|2x^2+x+3|+C$.